Learning to Trust Sparse Metric Evidence
for Monocular Depth Foundation Models

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Abstract

Monocular depth foundation models supply a strong dense geometric prior, yet metric scale and local depth can still be wrong. We ask how little external metric evidence is enough to ground that prior, in the regime of one to a handful of sparse observations. A few reliable measurements already yield large metric gains, but the same scarcity makes each measurement highly leveraged, so an unreliable point can erase the gain. Reliable Few-Point Metric Grounding (RFMG) is a lightweight module, trained while the backbone stays frozen, that learns which observations to trust and how much correction the reliable evidence may induce. Scarce reliable evidence is used mainly for global scale. Spatially varying refinement is enabled only as reliable evidence accumulates. On NYUv2 with frozen UniDepthV2, eight clean observations reduce absolute relative error from 0.0766 to 0.0540. When 80% of those observations are corrupted, RFMG still reaches 0.0608. Sparse depth is treated as compact metric evidence for a prior that already exists, rather than as incomplete geometry that must be densified from scratch.

1 Motivation and contributions

Monocular depth foundation models predict dense metric geometry from a single RGB image [1–3]. Dense metric sensing is often costly, while a lightweight sensor can supply only a few metric measurements, and those measurements are not uniformly reliable. The practical question is how little metric evidence is sufficient to ground the prior.

Depth completion fuses RGB with sparse depth to recover a dense map [4, 5]. Later methods use a pretrained monocular model as the geometric prior and let sparse measurements guide or adapt it [6–8]. Those designs are not built for the reliability problem that appears when only a handful of points are available. Each point then has unusual leverage, so a corrupted measurement can dominate the correction. Two decisions are coupled: which observations to trust, and how much correction the available evidence should be allowed to induce.

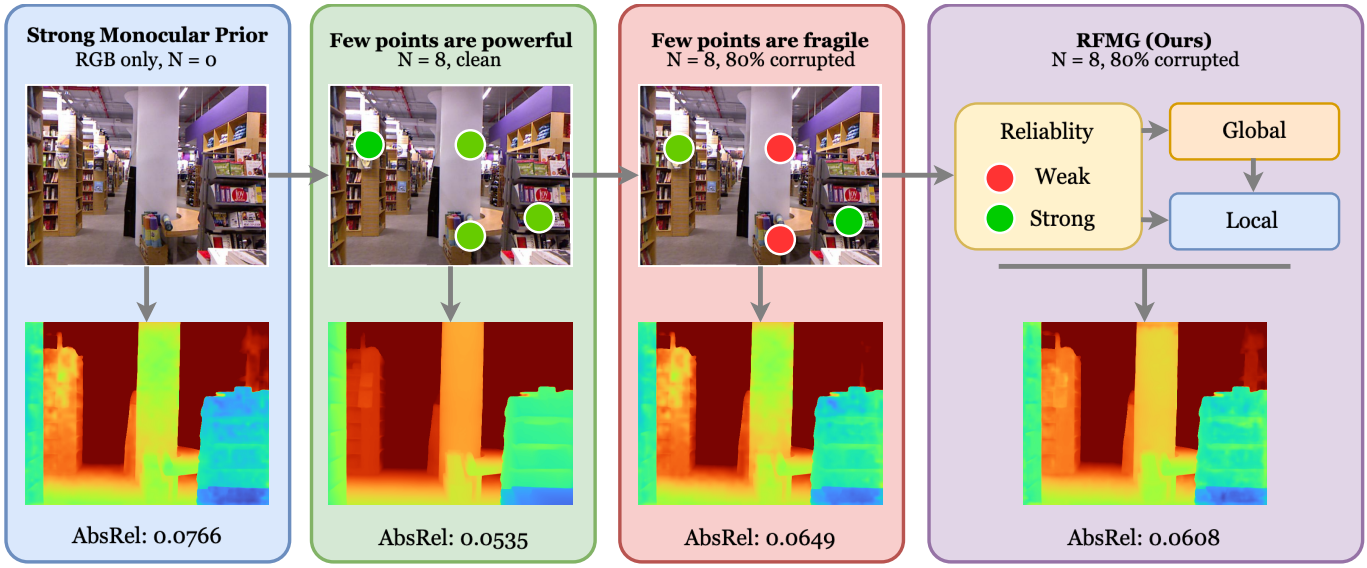


Figure 1: Few metric observations are powerful, and fragile. On NYUv2 with UniDepthV2, absolute relative error moves from 0.0766 for the RGB-only prior, to 0.0535 with direct local grounding on N=8 clean points, to 0.0649 when 80% of those points are corrupted, and to 0.0608 for RFMG under the same corruption. Marker colors indicate observation quality for visualization only.

We study $N \in \{1, \ldots, 64\}$ observations, with emphasis on N=1–16. The pattern is a power–fragility trade-off (Figure 1). When the prior already captures scene structure, a few metric points produce a large gain, and the same scarcity makes unreliable points consequential. Sparse measurements are used as compact metric evidence for an existing dense prior, not as an incomplete point cloud to be completed.

The contributions are as follows.

- Extreme few-point metric grounding of frozen monocular depth models, under scarce and possibly unreliable observations.
- RFMG, which separates observation reliability from correction authority: reliability-weighted global scale, then evidence-dependent local refinement.
- A check across three frozen backbones, four datasets, several budgets, and clean and corrupted measurements. Few clean points already help, and RFMG keeps more of that gain when many points are unreliable.

2 Method

An RGB image I and a sparse set $\mathcal{P} = \{(\mathbf{u}_i, d_i)\}_{i=1}^{N}$ are given. The goal is dense metric depth $\widehat{Z}$. The depth d_i may be corrupted. A frozen monocular model returns dense features F, an initial metric depth Z_0, and a backbone uncertainty cue C. RFMG does not rebuild geometry from $\mathcal{P}$. It first scores each observation, uses the trusted points for a global scale, and turns on local refinement only as reliable evidence grows (Figure 2).

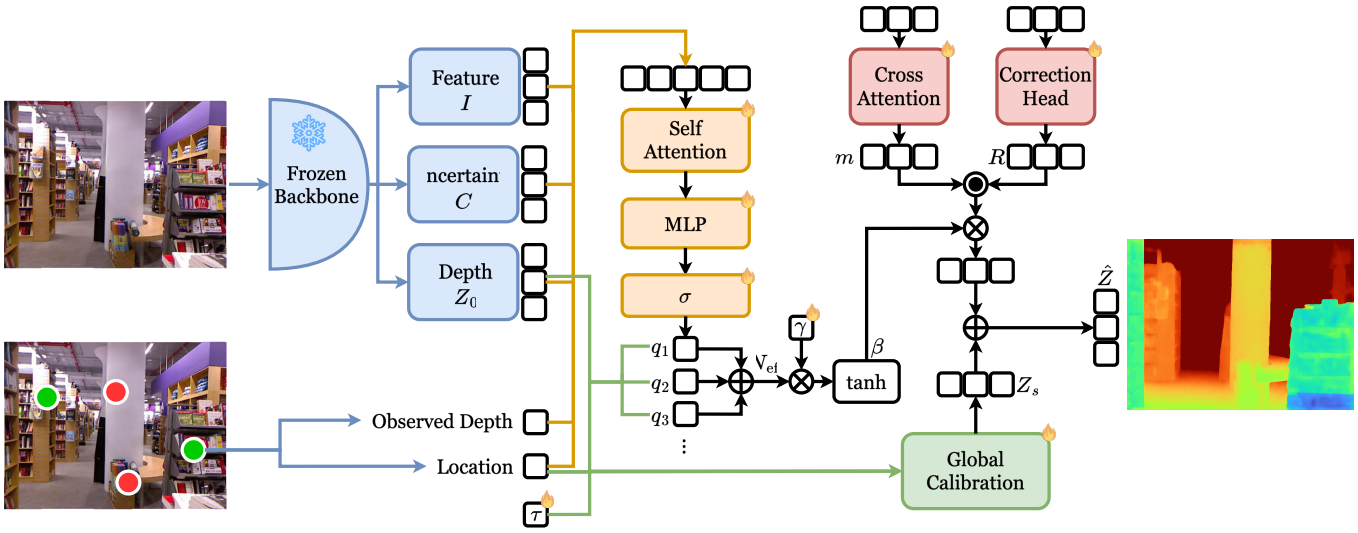


Figure 2: RFMG. A frozen backbone provides F, Z_0, and C. Self-attention contextualizes the sparse points and a reliability head predicts q_i. Reliability-weighted scale cues produce a calibrated depth Z_s. Cross-attention yields a support map m and a residual R. The effective reliable count N_{eff} sets the local authority β, and $\widehat{Z} = Z_s + \beta(m \odot R)$.

2.1 Which observations to trust

At each observation we sample $f_i = F(\mathbf{u}_i)$, $z_i = Z_0(\mathbf{u}_i)$, and $c_i = C(\mathbf{u}_i)$, and the log-depth residual $r_i = \log d_i - \log z_i$. The token is

$$x_i = [f_i, \mathrm{PE}(\mathbf{u}_i), \log d_i, \log z_i, r_i, c_i], \tag{1}$$

with PE a positional encoding of $\mathbf{u}_i$. A short self-attention encoder produces h_i, and a reliability head maps it to

$$q_i = \sigma(g(h_i)) \in [0, 1]. \tag{2}$$

The cue c_i is only an auxiliary input. The score q_i is what later gates both the global scale and the local update.

2.2 Global scale, then local refinement

Each point supplies a scale cue $t_i = d_i / z_i$. Reliability and a bounded transform $\widetilde{c}_i$ of the uncertainty cue give weights $w_i = q_i \widetilde{c}_i$, a total weight $W = \sum_i w_i$, and a weighted scale $\bar{t}$. Scarce evidence must not apply that scale in full, so the estimate is shrunk toward 1,

$$\widetilde{s} = 1 + \frac{W}{\tau + W}(\bar{t} - 1), \tag{3}$$

with learned $\tau > 0$. A bounded map of $\widetilde{s}$ is the scale s, and $Z_s = sZ_0$.

Local errors remain. Observations are re-encoded against Z_s and written into dense features by reliability-aware cross-attention. A null attention option lets the model abstain where the points do not support a local change. The assigned attention mass is a support map m, and a small head predicts a residual R. The amount of reliable evidence is the soft count $N_{\text{eff}} = \sum_i q_i$, which sets

$$\beta = \tanh\big(\gamma \max(N_{\text{eff}} - 1, 0)\big), \qquad \widehat{Z} = Z_s + \beta\,(m \odot R), \tag{4}$$

with learned $\gamma > 0$, so $\beta \in [0, 1)$. Thus q_i chooses which points count, m chooses where a local change is supported, R is the proposed residual, and β chooses how much of it is applied. When $N = 1$, $N_{\text{eff}} \le 1$ and $\beta = 0$, so RFMG reduces to global calibration. Local refinement appears only as further reliable points arrive.

2.3 Training

Only RFMG is trained. For each image, N is drawn uniformly from $\{1, \ldots, 16\}$ valid ground-truth locations. A corruption rate ρ is drawn from $(0, 0.8)$, and each point is perturbed independently with probability ρ. Clean points have reliability target 1 and corrupted points target 0. The loss supervises the final depth, the globally calibrated depth, and the reliability scores,

$$\mathcal{L} = \mathcal{D}(\widehat{Z}, Z^*) + \lambda_s\, \mathcal{D}(Z_s, Z^*) + \lambda_r\, \mathcal{L}_{\text{rel}}, \tag{5}$$

with $\lambda_s = 0.5$, $\lambda_r = 0.1$, binary cross-entropy for $\mathcal{L}_{\text{rel}}$, and $\mathcal{D}$ a sum of a log-depth term and an absolute metric term. The backbone stays frozen.

3 Results

NYUv2 [9] is the main benchmark (700/95/654 train/val/test images). ETH3D [10], KITTI [11], and DIODE [12] use a separate RFMG trained on that dataset, with the same frozen backbone. Points are sampled uniformly from valid ground-truth pixels. Training uses $N \in \{1, \ldots, 16\}$ and $\rho \in (0, 0.8)$. Evaluation reports absolute relative error (AbsRel) at $N \in \{1, 8, 16\}$ under clean, 40%, and 80% corruption. The backbones are Metric3D v2 ViT-L, UniDepthV2 ViT-L/14, and Depth Anything 3 (DA3) Metric-Large, all frozen. Compared methods see the same point coordinates.

Eight clean points already move the metric. With UniDepthV2, AbsRel falls from 0.0766 at N=0 to 0.0689, 0.0540, and 0.0401 at N=1, 8, and 16 (Table 1). DA3 moves from 0.0912 to 0.0831, 0.0545, and 0.0448. Metric3D v2 does not benefit from a single point (0.1257 to 0.1289) and does benefit once eight points are available (0.0677, then 0.0482 at N=16). One point is therefore not enough for every backbone. A few points are.

The same budget is fragile, and RFMG limits the damage. At N=8 on UniDepthV2, AbsRel rises from 0.0540 clean to 0.0563 at 40% corruption and 0.0608 at 80%, still well below the RGB-only 0.0766. DA3 rises from 0.0545 to 0.0597 at 80%. All three RFMG models stay ahead of their frozen RGB prior at N=8 and N=16 even when 80% of the points are corrupted. LDCM is the strongest clean baseline at those budgets (0.0488 and 0.0338) and loses more under corruption (0.0740 and 0.0569). At N=1, completion-style baselines remain far from the prior, while UniDepthV2+RFMG is at 0.0689 clean and 0.0767 at 80% corruption, essentially the RGB error when the single point cannot be trusted. Most of the gain on a fixed UniDepthV2+RFMG model appears by N=8. At N=1 the full model matches global calibration, because $\beta = 0$. Further points pull the full model ahead of global scale alone, under both clean and 80% corrupted inputs.

Table 1: NYUv2 AbsRel↓. For $N > 0$, each cell is Clean | 80% corruption. RGB-only methods have no sparse input.

Method	N=0	N=8	N=16
Metric3D v2	0.1257	–	–
UniDepthV2	0.0766	–	–
DA3	0.0912	–	–
PriorDA	–	0.0608 — 0.0818	0.0381 — 0.0598
LDCM	–	**0.0488** — 0.0740	**0.0338** — 0.0569
Metric3D v2 + RFMG	–	0.0677 — 0.0732	0.0482 — 0.0538
UniDepthV2 + RFMG	–	0.0540 — **0.0608**	0.0401 — **0.0442**
DA3 + RFMG	–	0.0545 — **0.0597**	0.0448 — **0.0505**

The pattern is not specific to NYUv2 (Table 2). With a frozen UniDepthV2 prior and an in-domain RFMG, AbsRel falls with N on ETH3D, KITTI, and DIODE as well. At N=8 and 80% corruption, every dataset remains better than its RGB-only prior. At N=1 and 80% corruption, NYUv2 returns to 0.0767, against 0.0766 with no points.

Table 2: UniDepthV2+RFMG across datasets, AbsRel↓. Each RFMG cell is Clean | 80% corruption at N=8. The backbone is frozen. A separate refiner is trained per dataset.

Dataset	RGB-only	RFMG, N=8
NYUv2	0.0766	0.0540 — 0.0608
ETH3D	0.2166	0.0509 — 0.0615
KITTI	0.1068	0.0562 — 0.0654
DIODE	0.6887	0.4269 — 0.4820

On NYUv2 with UniDepthV2 and N=8, direct local grounding is slightly best when every point is clean (0.0535) and rises to 0.0649 at 80% corruption. Full RFMG is 0.0540 clean and 0.0608 corrupted. Dropping the reliability head raises the three conditions from 0.0540/0.0563/0.0608 to 0.0588/0.0621/0.0653. Global calibration alone, with reliability but no local residual, sits at 0.0582/0.0601/0.0634. Reliability decides which evidence counts. The amount of reliable evidence decides how much local change that evidence may make.

4 Conclusion

A few reliable metric points can ground a frozen monocular depth prior, and the same scarcity makes an unreliable point dangerous. RFMG scores each observation and spends scarce trusted evidence on global scale, opening local refinement only as the trusted count grows. Across three frozen backbones and four datasets, the module keeps a large share of the few-point gain when most of the points are corrupted. Sparse measurements then do not have to reconstruct the scene. They can act as compact metric evidence for a dense prior that is already there.

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